

Information & Communication Technology

Complete Study Guide: Number Systems & Digital Representation

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1. Classification of Number Systems

A **Number System** is a system of expressing numbers. Modern computing uses several positional number systems based on different radices (bases).

Number System	Base / Radix	Symbols / Digits Used
Binary	2	0, 1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A (10), B (11), C (12), D (13), E (14), F (15)

2. Conversions Between Number Systems

A. Conversions from Decimal System

1. Decimal to Binary

1. Divide the decimal number repeatedly by 2.
2. Write down the quotient and record the remainder (0 or 1) on the right side.
3. Continue dividing until the quotient becomes 0.
4. Read the remainders from bottom to top (Most Significant Bit to Least Significant Bit).

Example: Convert 12_{10} to Binary

2	12	Remainder: 0
2	6	Remainder: 0
2	3	Remainder: 1
2	1	Remainder: 1
	0	

Reading bottom-to-top: $12_{10} = 1100_2$

2. Decimal to Octal

1. Divide the given decimal integer by 8.
2. Write down the remainder (0 to 7) to the right.

3. Continue dividing the quotient by 8 until it reaches 0.
4. Read the remainders from bottom to top.

Example: Convert 245_{10} to Octal

8	245	Remainder: 5
8	30	Remainder: 6
8	3	Remainder: 3
	0	

Reading bottom-to-top: $245_{10} = 365_8$

3. Decimal to Hexadecimal

1. Divide the decimal number by 16 as an integer division.
2. Record the remainder. If the remainder is greater than 9, convert it to its equivalent letter (10=A, 11=B, 12=C, 13=D, 14=E, 15=F).
3. Repeat until the quotient is 0.
4. Read remainders from bottom to top.

Example: Convert 3240_{10} to Hexadecimal

16	3240	Remainder: 8
16	202	Remainder: 10 (A)
16	12	Remainder: 12 (C)
	0	

Reading bottom-to-top: $3240_{10} = CA8_{16}$

4. Fractional Decimal to Binary

1. Multiply the fractional part by 2.
2. Record the integer part (0 or 1) before the decimal point.
3. Multiply the remaining fractional part by 2 again.
4. Repeat until the fractional part becomes 0 (or desired precision is met).
5. Read the integer parts from top to bottom.

Example: Convert 0.3125_{10} to Binary

$$0.3125 \times 2 = 0.625 \rightarrow \text{Integer part: } 0$$

$$0.625 \times 2 = 1.25 \rightarrow \text{Integer part: } 1$$

$$0.25 \times 2 = 0.50 \rightarrow \text{Integer part: } 0$$

$$0.50 \times 2 = 1.00 \rightarrow \text{Integer part: } 1$$

$$\text{Reading top-to-bottom: } 0.3125_{10} = 0.0101_2$$

5. Fractional Decimal to Octal

Example: Convert 0.3125_{10} to Octal

$$0.3125 \times 8 = 2.50 \rightarrow \text{Integer part: } 2$$

$$0.50 \times 8 = 4.00 \rightarrow \text{Integer part: } 4$$

$$\text{Reading top-to-bottom: } 0.3125_{10} = 0.24_8$$

B. Conversions to Decimal System

1. Binary to Decimal

Example: Convert 1101_2 to Decimal

Positional values from right to left: $2^3, 2^2, 2^1, 2^0$

$$1101_2 = (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$= (1 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1) = 8 + 4 + 0 + 1 = 13_{10}$$

2. Octal to Decimal

Example: Convert 1260_8 to Decimal

$$1260_8 = (1 \times 8^3) + (2 \times 8^2) + (6 \times 8^1) + (0 \times 8^0)$$

$$= (1 \times 512) + (2 \times 64) + (6 \times 8) + (0 \times 1) = 512 + 128 + 48 + 0 = 688_{10}$$

3. Hexadecimal to Decimal

Example: Convert $A0B1_{16}$ to Decimal

$$\begin{aligned} A0B1_{16} &= (10 \times 16^3) + (0 \times 16^2) + (11 \times 16^1) + (1 \times 16^0) \\ &= (10 \times 4096) + (0 \times 256) + (11 \times 16) + (1 \times 1) = 40960 + 0 + 176 + 1 = 41137_{10} \end{aligned}$$

C. Direct Conversions between Binary, Octal, & Hexadecimal

1. Binary to Octal

Group binary digits into sets of 3 from right to left. Add leading zeros if necessary, then convert each group to its octal equivalent.

Example: Convert 10011011_2 to Octal

Grouping into 3s: $10, 011, 011_2$

Adding leading zeros: $010, 011, 011_2$

Values: $(0 \times 4 + 1 \times 2 + 0 \times 1), (0 \times 4 + 1 \times 2 + 1 \times 1), (0 \times 4 + 1 \times 2 + 1 \times 1) = 2, 3, 3$

Result: 233_8

2. Binary to Hexadecimal

Group binary digits into sets of 4 from right to left. Add leading zeros if needed, then convert each group to its hexadecimal equivalent.

Example: Convert 11101100101001_2 to Hexadecimal

Grouping into 4s: $11, 1011, 0010, 1001 \rightarrow 0011, 1011, 0010, 1001$

Calculations: 3, 11 (B), 2, 9

Result: $3B29_{16}$

3. Octal / Hexadecimal to Binary

Convert each digit directly into its 3-bit binary equivalent (for Octal) or 4-bit binary equivalent (for Hexadecimal), then combine and remove non-significant leading zeros.

Examples:

Octal to Binary: $0137_8 = 000\ 001\ 011\ 111_2 = 101111_2$

Hexadecimal to Binary: $1A90_{16} = 0001\ 1010\ 1001\ 0000_2 = 1101010010000_2$

3. Representation of Signed Numbers

Computers represent negative numbers using specific conventions in fixed bit widths (typically 8-bit):

1. Sign-Magnitude Representation

The Most Significant Bit (MSB) represents the sign (**0 for positive, 1 for negative**), while the remaining 7 bits represent the magnitude. This method creates two representations for zero (+0 and -0).

- $+45_{10}$ in 8-bit Sign-Magnitude: 00101101_2
- -45_{10} in 8-bit Sign-Magnitude: 10101101_2

2. One's Complement

Positive numbers are represented as standard binary. For negative numbers, invert all bits (replace 0s with 1s and 1s with 0s).

- $+12_{10} = 00001100_2$
- -12_{10} (1's complement) = 11110011_2
- Has two representations of zero: $+0 = 00000000_2$ and $-0 = 11111111_2$.

3. Two's Complement

To find the 2's complement of a negative number, first take the 1's complement and then add 1. This system eliminates negative zero and allows standard binary addition circuits to perform subtraction.

Example: Represent -5_{10} in 2's Complement (8-bit)

1. Start with $+5_{10}$: 0000101_2
2. Take 1's Complement (invert bits): 1111010_2
3. Add 1: $1111010 + 1 = 1111011_2$ (in Hexadecimal: FB_{16})

Comparison of Representation Schemes

Method	Characteristics & Applications
Sign-Magnitude	Used only when no addition/subtraction is performed. Applied in Analog-to-Digital Converters (ADC).
One's Complement	Simpler hardware design due to straightforward logical inversion, but limited due to double-zero issue.
Two's Complement	Widely used in modern arithmetic units. Allows low-cost, high-speed hardware design for addition and subtraction.

4. Most & Least Significant Digits (MSD / LSD)

- **MSD (Most Significant Digit):** The digit with the highest positional value in a number.

- **LSD (Least Significant Digit):** The digit with the lowest positional value in a number.

Number	MSD	LSD
2975.0	2	5
56.034	5	4
0.03145	3	5
0031.0060	3	6

5. Character Encodings in Computers

Computers represent text, numbers, and symbols using standard character codes:

- **BCD (Binary Coded Decimal):** Encodes each decimal digit with a 4-bit binary sequence.
- **EBCDIC (Extended Binary Coded Decimal Interchange Code):** 8-bit character encoding used mainly on IBM mainframe systems.
- **ASCII (American Standard Code for Information Interchange):** 7-bit / 8-bit character set used broadly across microcomputers.
- **UNICODE:** Comprehensive variable-length encoding scheme capable of representing text across all human languages and character sets.

6. Binary Arithmetic

A. Binary Addition

Example: $101100_2 + 1100_2$

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101100_2
+  01100_2
-----
111000_2

```

Result: 111000_2

B. Binary Subtraction

Example: $101100_2 - 1101_2$

```
  101100_2
-   001101_2
-----
  011111_2
```

Result: 011111_2

7. Bitwise Operations

NOT Gate (Inversion)

A	NOT A
0	1
1	0

Example: $NOT\ 0111_2\ (7_{10}) = 1000_2\ (8_{10})$

AND Gate

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Example: $0101_2\ (5_{10})\ AND\ 0011_2\ (3_{10}) = 0001_2\ (1_{10})$

OR Gate

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

Example: $0101_2 (5_{10}) \text{ OR } 0011_2 (3_{10}) = 0111_2 (7_{10})$

XOR Gate (Exclusive OR)

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

Example: $0010_2 (2_{10}) \text{ XOR } 1010_2 (10_{10}) = 1000_2 (8_{10})$

Reference Source: ICT Teacher Guide Handbook | Compiled & Translated by Tech Hub